ZZ-G-401B
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SUPERSEDING
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November 27, 1963

FEDERAL SPECIFICATION

GLOVES, RUBBER (FOR) ELECTRICAL WORKERS (FOR USE IN CONNECTION WITH APPARATUS OR CIRCUITS NOT EXCEEDING 5,000 VOLTS TO GROUND)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers seamless-type rubber insulating gloves of one grade, used for the protection of electrical workers from electric shock while working on energized conductors and equipment. Caution: The proof test voltage referred to in this specification should not be construed to mean the safe voltage on which the gloves can be used (see 6.1).

1.2 Classification. The gloves shall be of the following classes, sizes and lengths, as specified (see 6.2):

Class 0. - Proof test 5,000 volts, 3 minutes (maximum safe voltage - 750).

Class 1. - Proof test 10,000 volts, 3 minutes (maximum safe voltage - 3,000).

Class 2. - Proof test 15,000 volts, 3 minutes (maximum safe voltage - 4,000).

Class 3. - Proof test 20,000 volts, 3 minutes (maximum safe voltage - 5,000).

Sizes: 9, 9-1/2, 10, 10-1/2, 11, 11-1/2, and 12.

Lengths: Class 0 only - 10-1/2 inches.
Classes 1, 2, and 3: - 14, 16 and 18 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Fiberboard.
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 601 - Rubber: Sampling and Testing.

(Activites outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Services Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

FSC 8415

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply:

American Society For Testing And Materials (ASTM) Standards:

ASTM-D-120-68 - Rubber Insulating Gloves.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Attention: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material. The gloves shall be made of high-grade natural rubber or rubber compound, pigmented black, suitably compounded and vulcanized to meet the requirements in 3.4 thru 3.8.2.

3.2 Construction. The gloves shall be made by the "dipping" or other seamless process using the rubber specified in 3.1. Construction shall conform to 3.2.1 and 3.2.2.

3.2.1 Cuff edges. Unless otherwise specified, the gloves shall be rolled outward along the top edge to form a self binding or the gloves shall be left flat and reinforced along the top edge with a band of rubber of the same composition as the rubber in the glove. When the glove edge is rolled, the diameter of the roll shall be 3/16 ± 1/16 inch for the class 0 and 1/4 ± 1/16 inch for the class 1, 2 and 3 gloves. When a reinforcement band is used, the band shall be approximately 1/4 inch thick and approximately 1/4 inch wide.

3.2.2 Fingers. The fingers of the gloves shall conform with the natural shape of the human hand insofar as possible and shall be of the curved finger style. The circumference of the fingers shall be such as to prevent binding and to afford comfort to the user. (Comfort may be judged by the feel of the glove on the hand in closing the hand). The tip of the thumb shall turn inward slightly toward the second finger or the thumb shall be straight. The base of the thumb shall be placed in front of the first finger, with the base set out as it is on the natural hand. The base of the thumb on the palm side shall be continued to the glove palm in the position specified above with a gentle eliptical curve, so as to keep the so-called thumb crotch area and part just below the thumb crotch area on the palm side without a noticeable crease. The absence of such a crease is especially noticeable when the glove is reversed.

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3.3 Dimensions.

3.3.1 Size. The size of each glove shall be as specified (6.2), and shall be expressed in inches as the distance around the palm at the base of the knuckles. The permissible variation in size shall be plus or minus 1/4 inch. (See 4.4.7.)

3.3.2 Length. The length of each glove shall be as specified (6.2), and shall be expressed in inches as the distance from the tip of the second finger to the outside edge of the cuff. The permissible variation in length shall be ± 1/2 inch.

3.3.3 Thickness. The thickness of rubber of the four classes of gloves shall be as specified in table I when determined as described in 4.4.1.

TABLE I. Thickness limits

Class	Crotch (min.)	Thickness other than crotch		
		Minimum	Area A* (max.)	Area B* (max.)
	Inch	Inch	Inch	Inch
0	0.018	0.021	0.035	0.040
1	0.030	0.035	0.050	0.065
2	0.038	0.045	0.060	0.080
3	0.050	0.060	0.085	0.100

* The parts of the glove designated as areas A and B for maximum thickness measurements are as follows: Area A extends from the ends of the thumb and fingers, excluding the tips, up to an imaginary line around the glove at the thumb crotch. Area B extends from the edge of the cuff down to the liner at the thumb crotch and also includes the thumb tip and finger tips. The thumb or finger tip is the area of rubber from the extreme tip of the finger to a line around each finger in a plane one-half inch distant axially from the tip (see 6.4).

3.4 Tensile strength and ultimate elongation. The tensile strength of the glove shall be not less than 2,500 pounds per square inch and the ultimate elongation shall be not less than 600 percent (1 inch stretched to 7 inches) when determined as specified in 4.2.2.

3.5 Set. The maximum set of the glove shall not exceed 0.25 inch after a stretch of 1 inch to 6 inces as specified in 4.2.2.

3.6 Adhesion between reinforcing band and top edge. If the top edge of the glove is reinforced with a strip of rubber as described in 3.2.1, the adhesion between the two layers shall be not less than 8 pounds per inch of width when tested as specified in 4.2.2.

3.7 Resistance to accelerated aging.

3.7.1 Resistance to air. After aging in air, the tensile strength and ultimate elongation shall be not less than 75 percent of the original value when tested as specified in 4.2.2.

3.7.2 Resistance to oxygen. After aging in oxygen, the tensile strength and ultimate elongation shall be not less than 75 percent of the original value when tested as specified in 4.2.2.

3.8 Electrical properties.

3.8.1 Proof test voltage and proof test current test (see 6.5 and 6.6). Each glove produced shall withstand the 60 cycles per second alternating voltage (root means square value) specified in table II, applied continuously for 3 minutes when tested as specified in 4.2.2. The 60 cycles per second proof test current shall not exceed the values specified in this table at any time during the 3-minute test when tested as specified in 4.2.2. If this test is made at any other frequency, the permissible proof test current shall be computed from the directed ratio of the frequencies (4.4.2).

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TABLE II

Class of glove	Three minute proof test voltage	Maximum proof test current r.m.s. milliamperes			
		Length of glove (inches)			
		10-1/2	14	16	18
	r.m.s volts				
0	5,000	8	-	-	-
1	10,000	-	10	12	14
2	15,000	-	12	14	16
3	20,000	-	14	16	18

3.8.2 Breakdown voltage. The gloves shall not break down at voltages below those specified in table III when tested as specified in 4.2.2. (See 6.7).

TABLE III

Class of glove	Water level distance from edge of glove, maximum	Breakdown voltage, minimum
	Inches	Volts
0	2	6,000
1	3	20,000
2	3-1/2	25,000
3	4	30,000

3.9 Marking. Each glove shall be marked with the name of the manufacturer, size of glove, class, length, date of manufacture, and the maximum voltage at which the glove is permitted to be used. Such marking shall be confined to the back of the glove within 2 inches of the cuff edge and shall be permanent, nonconducting, and so applied as not injure the glove. The color of the marking shall be white.

3.10 Workmanship. The finished gloves shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the specified quality levels (AQLs).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of end item.

4.2.1.1 Examination of end item. The defects found during examination of the end product shall be classified in accordance with paragraphs 4.2.1.1.1 and 4.2.1.1.2. The applicable inspection levels and acceptable quality levels (AQLs) shall be as indicated in paragraph 4.2.1.1.3. The lot size shall be expressed in terms of one glove. The sample unit for this examination shall be one completely fabricated glove and the selection shall be by pairs.

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4.2.1.1.1 Visual examination of gloves. The gloves shall be examined for defects in pairing, color, design, construction, material workmanship and marking. For the pairing examination, the pair shall be examined together. Each defect found during the examination for pairing shall be scored as a single defect.

Examine	Defect	Classification	
		Major	Minor
Pairing	Not properly mated, e.g., not right and left of same size and length.	X	
	Definite variation in color or appearance.		X
Cleanliness	Grease or oil stains or other dirt on outside or inside of glove which cannot be readily removed.	X	
	Not clean, i.e., dirty, but dirt can be removed with a cleaning agent.		X
Design	Any characteristic not in accordance with specified requirements (unless otherwise indicated herein).	X	
Color	Not black in color.		X
Construction and workmanship (applicable to inside and outside)	Not seamless construction.	X	
	Any cut, hole, tear, rip or rupture through the material.	X	
	Any closed blister.	X	
	Any burned spot which cracks on flexing, bending, or stretching (by hand).	X	
	Any pinch, pit, thin spot, abraded area, deep crease, readily removable foreign matter (see note below) or similar defective condition which results in a thickness at the defect less than the minimum thickness allowed for the glove.	X	
	Any one inch square area which contains more than three pinches, pits, thin spots, abraded areas, deep creases, readily removable foreign matter, (see not below) or similar defective condition not resulting in a thickness at the defect less than the thickness allowed for the glove.		X
	Any foreign matter not readily removable (see note below).		X
	NOTE: Readily removable foreign matter is defined as material which can be removed by flexing or stretching the glove by hand or by rubbing the defective area with the ball of the thumb or fingers. Cutting, scratching, or otherwise puncturing the surface is not permitted.		

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Examine	Defect	Classification	
		Major	Minor
Construction	Any repair or patch.	X	
	Any solid rubber ridge, run or lump resulting in a thickness at the ridge, run or lump greater than the maximum thickness allowed for the glove.		X
	Evidence of improper vulcanization e.g., tacky, etc.	X	
	Any malformation or distortion seriously affecting service-ability.	X	
	Any malformation or distortion affecting serviceability but not seriously.		X
Marking	Manufacturer's name, date or manufacture, class, length, or size missing, incorrect, incomplete, illegible, not in specified location, not white or not accomplished as specified.		X
	Voltage rating omitted, incomplete or illegible.	X	
	Not permanent, i.e., can be easily rubbed off with moistened thumb.		X

4.2.1.1.2 Dimensional examination (including thickness). The gloves shall be examined for defects in dimensions including thickness. Any dimension or thickness that is not within the established tolerance shall be classified a defect. The sample unit for this examination shall be one finished glove.

4.2.1.1.3 Inspection levels and acceptable quality levels (AQLs). The applicable inspection levels and acceptable quality levels, expressed in defects per hundred units, shall be as follows:

Examination	Inspection level	AQL	
		Major	Total
Paragraph 4.2.1.1.1	II	1.5	4.0
Paragraph 4.2.1.1.2	S-3	---	4.0

4.2.2 Testing the end item. Each lot of the end item shall be tested for the characteristics shown in table IV. Each glove produced shall be tested for conformance with the proof test voltage and proof test current requirements and any glove failing to meet these requirements shall be rejected. For the break-down voltage characteristic and Government verification for proof test voltage and current characteristics, the inspection level shall be S-4. The sample unit shall be two gloves and any failure encountered shall be cause for rejection of the lot represented by the sample. For all other test characteristics, the inspection level shall be S-1 and an AQL of 4.0 shall be applicable. The sample unit for these tests shall be four gloves. The lot size for the purpose of determining the sample size in accordance with MIL-STD-105 shall be expressed in units of one glove each and the selection shall be made in pairs.

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TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method paragraph	No. of determinations per sample unit	Results reported as: Pass or Fail	Results reported as: Numerically to nearest:
Proof test voltage and proof test current	3.8.1	4.4.2	1	X	
Breakdown voltage	3.8.2	4.4.2	1	X	
Material	3.1	Standard commercial	1	X	
Tensile strength:					
Initial	3.4	4.4.3	3		10 p.s.i.
After aging in air	3.7.1	4.4.6.1	3		10 p.s.i.
After aging in oxygen	3.7.2	4.4.6.2	3		10 p.s.i.
Ultimate elongation:					
Initial	3.4	4.4.3	3		10 percent
After aging in air	3.7.1	4.4.6.1	3		10 percent
After aging in oxygen	3.7.2	4.4.6.2	3		10 percent
Set	3.5	4.4.4	1		0.01 inch
Adhesion of reinforcing band (when applicable)	3.2.1	4.4.5	3		0.5 lb. per inch of width
Thickness	3.3.3	4.4.1	9 min.		0.001 inch

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine whether packaging, packing and marking complies with Section 5 requirements of this specification. The sample unit shall be one shipping container, with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) expressed in defects per 100 units shall be 2.5.

Examine	Defect
Marking	Omitted, incorrect, illegible, improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged.
Workmanship	Inadequate application of components such as: Incomplete closure of container flaps, improper taping, loose strapping or inadequate stapling. Bulged or distorted container.
Weight	Weight of contents of shipping container exceeds 65 pounds.

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4.4 Tests.

4.4.1 Thickness measurements. Thickness measurements made on the finished gloves shall be made with a micrometer graduated to 0.001 having an anvil about 1/4 inch in diameter and a presser foot 0.125 ± 0.01 inch in diameter exerting a total force of 3.0 ± 0.1 oz. (see 6.4). The thickness of the glove shall be measured at not less than four points on the face and four points on the back. Measurements shall also be made at one or more points in the crotch of the thumb and index finger and in the crotches between the fingers.

4.4.2 Electrical tests.

4.4.2.1 Proof test voltage, proof test current, and breakdown voltage test procedures. The proof test voltage, proof test current, and breakdown voltage shall be performed by filling the test glove with, and immersing the test glove in fresh tap water (60° to 90°F.) to within 1-1/2 inches of the edge of the cuff for test voltages of 10,000 volts, to within 2 inches for voltage of 15,000 volts and to within 2-1/2 inches for voltages of 20,000 volts. This distance may be increased to the values in table III for breakdown voltage tests for voltages of 20,000 volts and above. The water levels in the inside and outside of the gloves shall be the same. The water inside the glove and that outside the glove form the test electrodes. These shall be connected to the testing transformer by means of a chain or a sliding rod dipping into the water in the glove and by direct connection to the metal containing vessel. The water shall be free of air bubbles and air pockets inside or outside the glove and the exposed portion of the glove above the water line shall be dry. If forms or partitions of metal are used for supporting the gloves in the test tank, the metal shall not project above the water line. The glove shall be right side out.

4.4.2.2 Transformer. The desired test voltage may be most readily obtained by a step-up transformer energized from an adjustable low-voltage source. The transformer and its controlling equipment shall be of such size and design that with the test specimen in circuit, the crest factor (ratio of maximum to mean effective) of the test voltage shall not differ by more than 5 percent from that of sinusoidal wave over the upper half of the range of test voltage.

4.4.2.2.1 The crest factor may be checked by the readings of a peak voltmeter connected in parallel with the regular r.m.s. voltmeter. Alternately, the high side crest voltage may be determined by using a standard sphere-gap at the same time the r.m.s. high voltage is measured with either an electrostatic voltmeter or an r.m.s. voltmeter connected on the low voltage side of an instrument voltage transformer for which the ratio is known.

4.4.2.2.2 The current rating for the control equipment and supply shall be adequate for the rating of the test transformer which shall have a rating of not less than one-half kv.-a per glove being simultaneously tested, i.e., for ten gloves at one time 5 kv.-a. The test transformer shall be not less than 2 kv.-a rating.

4.4.2.3 Circuit breaker. The test transformer circuit shall be protected by an automatic circuit-breaking device designed to open promptly on the current produced by breakdown of a glove under test.

4.4.2.4 Voltage control. Control of voltage may be secured in one of several ways: (1) general field regulation (usually preferable); (2) variable ratio autotransformer; (3) induction regulator. Preference should be given to equipment producing no voltage distortion and having an approximately straight-line voltage-time or voltage-displacement curve over the operating range. Motor drive for the regulating equipment is convenient and tends to give uniform rate of rise to the test voltage. The frequency of the source shall be not more than 65 cycles per second and the source together with control arrangements and the test transformer shall provide a test voltage of approximately sine wave form as covered under 4.4.2.2.

4.4.2.5 Voltage measurement. The voltage shall be measured by a method which gives correct root-mean-square (r.m.s.) values of the voltage actually applied to the gloves. Approved methods are: (1) a voltmeter used in conjunction with a calibrated instrument potential transformer connected directly across the high-voltage circuit; (2) a calibrated electrostatic voltmeter connected directly across the high-tension circuit; (3) a voltmeter connected to a well-designed tertiary coil in the test transformer, provided it has been demonstrated that the assigned ratio of transformation does not change appreciably with load.

A voltmeter connected to the low-voltage side of the testing transformer may be used only if the ratio of transformation has been properly determined and is known not to change appreciably with load. A calibrated sphere-gap may be used to check the accuracy of the voltage as indicated by the voltmeter.

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4.4.2.6 Over-voltage sphere-gap protection. When making proof test voltage and proof test current tests, the test equipment shall have a spark gap connected in parallel with the gloves under test across the high-voltage circuit for the purpose of protecting against over-voltages. The spark gap shall be set to flash over a voltage of not more than 500 volts above the test voltage to be applied. The correct gap setting shall be determined on the basis of actual test with no gloves in circuit, by comparison with the voltage-measuring equipment. If test is to be continued after completion of the proof voltage test to determine break-down in voltage, the protective sphere-gap should then be removed from the circuit.

4.4.2.7 Application of proof test voltage. The proof test voltage shall be applied at a low value and gradually and steadily raised at a rate of approximately 1000 v. per sec. until the prescribed proof test voltage is reached (or failure occurs). The test period shall be counted from the instant when the prescribed testing voltage is reached. At the end of the test period the applied voltage shall be reduced to at least half-value (unless failure has occurred), before opening the test circuit.

4.4.2.8 Application of breakdown test voltage. The breakdown test voltage shall be applied at a low value and gradually and steadily raised at a rate of approximately 1000 v. per sec. until breakdown occurs or until the specified maximum breakdown voltage has been reached or exceeded. The maximum voltage observed before breakdown shall be recorded. When the specified breakdown voltage has been reached, it shall be held for more than 5 seconds before gradually and steadily reducing at a rate of approximately 1000 v. per sec. until zero voltage is reached.

4.4.2.9 Proof test current measurement. The proof test current shall be measured directly, by inserting a milliameter in series with each individual glove in turn. The reading should preferably be taken near the end of the proof voltage test period. It is customary to make high-voltage tests of this character with one end of the circuit grounded. When proof test current tests are made on one glove at a time, the metal containing vessel usually is made the grounded end of the circuit. The milliameter is connected in the grounded end of the circuit and shunted by a short-circuiting automatic self-closing switch which keeps the circuit closed except at the instant of reading and thus maintains an uninterrupted ground. When more than one glove is to be tested for proof test current at one time, the metal-containing vessel must be at high potential if it is required that the water electrodes inside the gloves shall be the ground electrode. The ammeter for reading proof test current is then connected in the ground electrode circuit through a suitable switching arrangement to permit reading the proof test current in each glove separately. If the ammeter and switching arrangement are suitably insulated, they may be used in the high-voltage circuits of the gloves, and the metal-containing vessel may be grounded, which is often more convenient.

4.4.3 Tensile strength and ultimate elongation. The tensile strength and ultimate elongation of the glove shall be determined in accordance with methods 4111 and 4121, respectively of Fed. Std. No. 601.

4.4.4 Set. The maximum set of the glove shall be determined in accordance with method 4411 of Fed. Std. No. 601.

4.4.5 Adhesion of reinforcing band to top edge. If the glove is reinforced along the top edge by a band of rubber as described in 3.2.1 the adhesion between the two layers shall be determined in accordance with method 8011 of Fed. Std. No. 601, using a straight specimen.

4.4.6 Resistance to accelerated aging.

4.4.6.1 Resistance to air. The resistance of the glove to air shall be determined in accordance with method 7221 or 7231 of Fed. Std. No. 601. The deterioration shall be measured by the change in tensile strength and elongation.

4.4.6.2 Resistance to oxygen. The resistance of the glove to oxygen shall be determined in accordance with method 7111 of Fed. Std. No. 601. The deterioration shall be measured by the change in tensile strength and elongation.

4.4.7 Size, outer circumference. The size or outer circumference of the glove as measured around the palm at the base of the knuckle shall be determined in accordance with method 2331 of Fed. Std. No. 601.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A. Each pair of gloves shall be packaged in a snug-fitting folding paperboard box conforming to variety 1, style III, type G, class i of PPP-B-566; or setup paperboard box conforming to type I variety 1, class A or C, style 4 of PPP-B-676. Each box shall be securely closed with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45, applied at the center of the length opening along the bottom and up each side a minimum of one-half of the depth dimension of the box.

5.1.2 Level C. Each pair of gloves shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Gloves of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B. Gloves of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification.

5.2.3 Level C. Gloves, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Precautionary marking. The following precautionary markings shall be applied in a conspicuous manner on all interior packages:

"CAUTION

These gloves are for use on electrical apparatus or circuits not exceeding 5000 volts. To prevent serious injury or death, please observe the following storage regulations:

a. Gloves shall be stored in original boxes right side out, not folded.

b. Do not store in direct sunlight or near steam pipes, radiators, or other artificial heat, or in a room where ambient storage temperature exceeds 90°F. (32.2°C.).

c. Do not expose gloves to oils, greases, or other foreign matter.

d. Gloves in storage for more than 6 months shall be carefully examined and tested for proof test current in accordance with ZZ-G-401.

e. Gloves in service shall be retested after each month of service."

5.3.4 Special marking. Each shipping container shall have the top and one side panel marked by stenciling or printing with black ink the following legend:

"STORE IN A COOL DRY PLACE"

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6. NOTES

6.1 Intended use. The rubber gloves covered by this specification are intended for use by electrical workers in the following applications:

Class 0 - For use only in connection with apparatus or circuits not exceeding 750 volts to ground.

Class 1 - For use only in connection with apparatus or circuits not exceeding 3,000 volts to ground.

Class 2 - For use only in connection with apparatus or circuits not exceeding 4,000 volts to ground.

Class 3 - For use only in connection with apparatus or circuits not exceeding 5,000 volts to ground.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class, size,and length required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Designation. The technical requirements of this specification are essentially the same as those of Specification D120-68 adopted by the American Society for Testing and Materials.

6.4 Thickness measurement. Areas A and B described in 3.3.3 are illustrated in Figure 1 of ASTM-D120-68. A convenient apparatus set-up for determining glove thickness in accordance with 4.4.1 is illustrated in Figures 2 and 3 of ASTM-D120-68.

6.5 The "Proof test" voltage referred to in this specification should not be construed to mean the safe voltage on which the glove can be used. The maximum voltage on which gloves can safely be used depends on many factors including the care followed in handling, storing, and inspecting the gloves in the field, the routine established for periodic laboratory inspection and test, the quality and thickness of the rubber and the design of the gloves together with such other factors as age, usage, weather conditions, etc.

6.6 It is common practice for the user of rubber protective equipment to prepare complete instructions and regulations to govern in detail the correct and safe use of such equipment. These should include provision for proper-fitting leather protective gloves needed to protect rubber gloves against cuts and snags. Inasmuch as gloves are used for personal protection and a serious personal injury or death may result if they fail while in use, an adequate factor of safety should be provided between the maximum voltage on which they are permitted to be used and the voltage at which they are tested. Gloves should not be exposed to oils, greases, or foreign matter of any kind.

6.7 It is essential that gloves which have been subjected to the breakdown voltage test should not thereafter be used for electrical protection, unless on subsequent proof test the proof test current is found not to be larger than on the initial proof test.

6.8 Gloves in storage and service. Gloves which have been in storage more than 6 months should be carefully examined and tested for proof test current in accordance with this specification. Gloves in service should be retested after each month of service. It is important that every glove be tested, not just certain percentages, because one defective glove may either cause death or serious injury to the person using it on electrical apparatus or circuits.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION (Gloves, Rubber (For) Electrical Workers (For Use In Connection With Apparatus or Circuits Not Exceeding 5,000 Volts To Ground) ZZ-G-401B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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